

Coding & Solution

Date	01 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/snehavagu-source/Voice-Based-Concept-Understanding-Analyser-.git
Programming Language(s)	Python
Framework(s) Used	FastAPI
Key Features Implemented	Audio Recording & Processing, Speech Transcription, AI Evaluation Engine , Fluency Metrics Tracking , Visual Analytics, Automated Reporting, Local Storage.
Pending / Incomplete Features	Real-time live audio plotting (currently waveform is generated after submission), and user role-based access control (Student vs Evaluator view partition).
Setup / Run Instructions	Install Dependencies, System Requirement, Configuration, Start Server, Access App

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

Modular Code Structure: The codebase is split logically into sections including Database Setup, Audio-to-Text conversion, Gemini API orchestration, and PDF Document building.

API Safety Catch: Implemented strong fallback exceptions so that even if the Gemini API fails or the network drops, the application smoothly degrades to local mode instead of crashing.